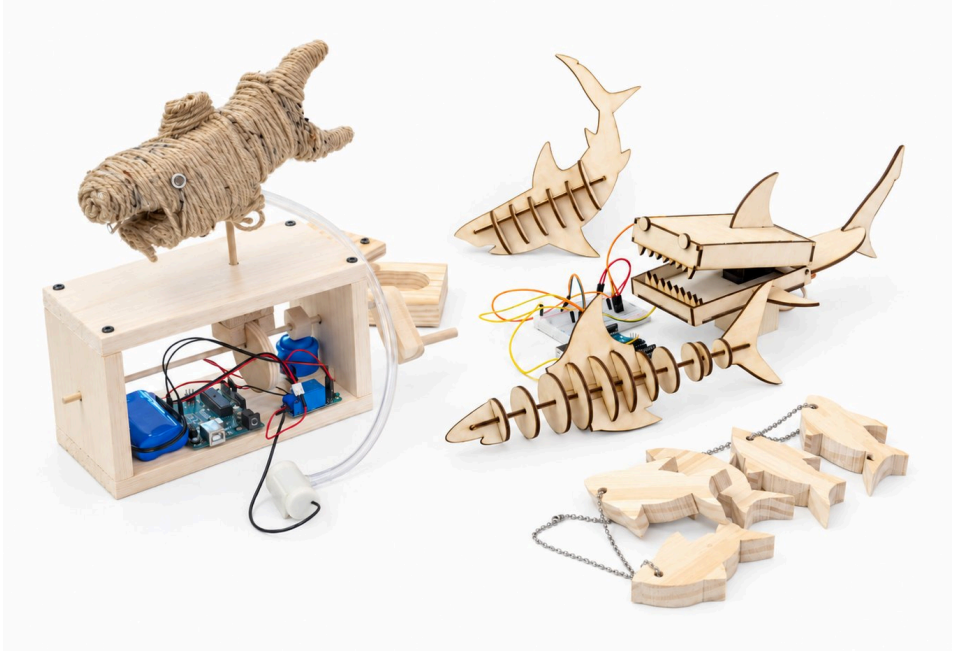
SHARKS, AGAIN.



Everything from five weeks of shop time, in one place.

My theme is sharks, so I made sharks again. And again. Five objects from one semester of shop time: a box that holds a key, five charms on a chain, a slotted skeleton at two scales, a servo head that bites at my hand, and a crank that makes a shark spit water.

COURSE	Intro to Fabrication
SPAN	Five weeks, five objects
SHOP	Band saw · drill press · laser cutter · sander
CONTROL	Arduino · servos · MediaPipe hand tracking
JOURNAL	Full process notes ↗

THE CLIPS IN THIS PORTFOLIO ARE MOVING IMAGES. WATCH THEM AT POMIESTONE-DEL.GITHUB.IO/SHARKS-AGAIN

INDEX

01	**Box Making**	*Wood, cut, drilled and hollowed*	WEEK 01
02	**Five Things**	*Five sharks on a single chain*	WEEK 02
03	**Big & Small**	*Slotted skeletons at two scales*	WEEK 03
04	**Enclosure**	*A head that opens its mouth*	WEEK 04
05	**Shark Automaton**	*Final project — hand-cranked, sprays water*	WEEK 05

Box Making



The lid pivots on a single screw over a round recess.

Trim the edges square, drill the recess on the drill press, separate the lid from the body, round the edges, hollow it out, then add a screw so the lid swings. My apartment key fits inside exactly.

WHAT WENT WRONG

This was the second box. On the first one I started cutting before drilling the holes, then rushed the recess deeper and went straight through the bottom.

TOOLS	Band saw, drill press, Forstner bit, sander
MATERIAL	Wood

Five Things



Five charms, one chain, each one cut and sanded by hand.

My theme is sharks, so five of them went onto one chain. References off Pinterest, a rough Blender mockup to picture the set, then outlines drawn straight onto the board, cut on the band saw and clamped for drilling.

THE HARD PART

Sanding. The tight inside corners between fin and body were difficult to reach, and a few of them never came out completely clean.

TOOLS	Blender, band saw, sander, drill press, clamp
MATERIAL	Wood, metal ball chain

Big & Small



The same structure twice: plywood at full size, 3 mm acrylic in miniature.

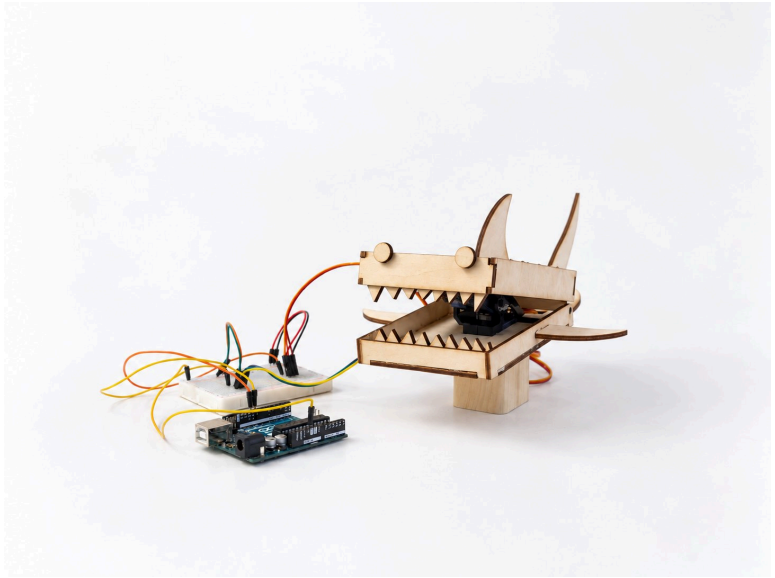
A sliced-structure project from MIT CBA gave me the logic. I retraced the file in Illustrator and reset every slot to 3 mm to see how it went together. Then I drew my own shark: the silhouette split lengthwise into four parts in Figma, circular ribs drawn in Illustrator, everything laser cut and slotted by hand.

WHAT WENT WRONG

The first acrylic cut had the dimensions wrong, so nothing slotted together and one rib snapped. The second version holds — with a little glue.

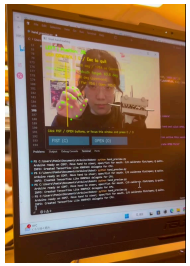
TOOLS	Illustrator, Figma, laser cutter
MATERIAL	Plywood, 3 mm acrylic
REFERENCE	MIT CBA, sliced structure ↗

Enclosure



The upper shell lifts on a servo; the teeth are offset so the mouth can close.





STILLS — THE CLIPS ARE MOVING ON THE SITE

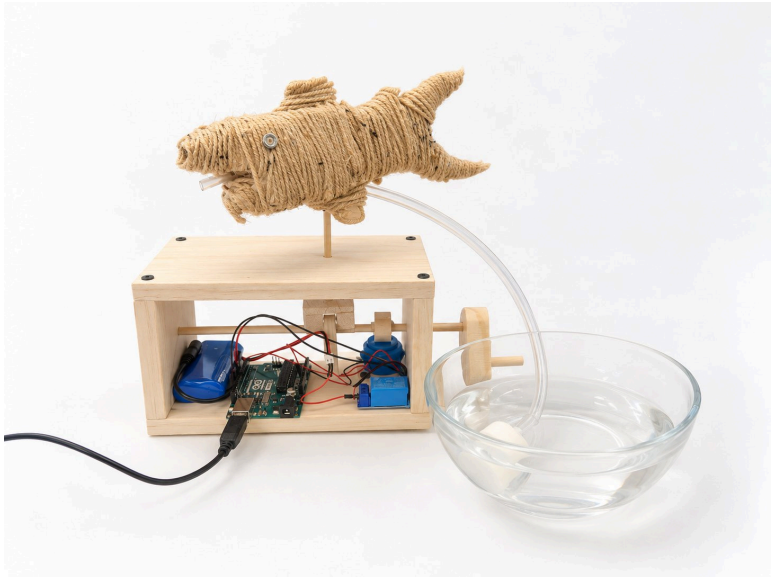
An enclosure for two servos and a pan-tilt bracket I carried over from home. Keeping the shark's curves in laser-cut plywood was too hard, so the head went boxy — which let one servo lift the whole upper shell to open the mouth. A webcam tracks my hand through MediaPipe: the head turns toward it, and the mouth closes when I close my fist.

WHAT WENT WRONG

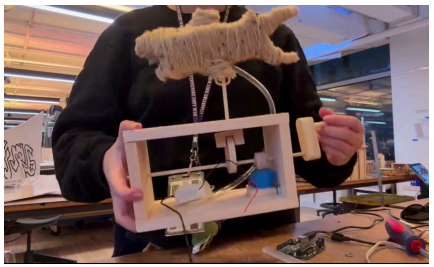
Three boxes before it fit. The first ignored the shape of the servo; the second had so much kerf compensation the pieces wouldn't go together at all. And the mouth servo I blamed for a week was fine — the supply voltage was too low. A 5 V lithium pack fixed it.

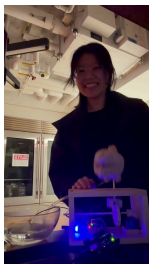
TOOLS	Flexcut3D, laser cutter, Arduino, MediaPipe
MATERIAL	Plywood, two servos, pan-tilt bracket, wooden base

Shark Automaton



Yarn-wrapped body, wooden crank, and a bowl to catch what comes out.





STILLS — THE CLIPS ARE MOVING ON THE SITE

The final project: a hand-cranked shark that sprays water when a cam inside comes around and presses a button. It began as a cardboard prototype — an arcade button turned out to be the only one easy enough to trigger — then went through several Blender models before anything was cut. The wood I ordered turned out to be balsa, so light and soft that screws went straight in without pre-drilling. The shark itself is carved foam; the surface came out rough, so I wrapped it in yarn I found in a cabinet behind me.

THE HARD PART

Measuring. Every part of the mechanism was tied to every other one, so changing a single dimension meant changing all the rest — and I kept forgetting. I lost count of how many times I thought it just wasn't going to work. It used almost every tool I had learned in the class; it felt a bit like a season finale.

LOOKING BACK

Before this class I never imagined I could build a little contraption like this. I was scared of a lot of new things, and got through them one small step at a time. A lot of the work isn't perfect, but the time I spent making it was unforgettable.

TOOLS	Blender, screwdriver, Arduino, carving tools
MATERIAL	Balsa wood, carved foam, yarn, arcade button, pump and tube